

In this problem set, the questions marked with a $(\star)$ are required for students enrolled in 583 and bonus for students enrolled in 483.

1. Consider flat spacetime written in cylindrical coordinates

$$(t, r, \phi, z),$$

with line element

$$ds^2 = -dt^2 + dr^2 + r^2 d\phi^2 + dz^2.$$

- (a) (6 points) Coordinate Basis

1. Write down the coordinate basis vectors.
2. Compute the squared norm of each basis vector using the metric.
3. Explain why this basis is not orthonormal, even though the spacetime is flat.

Solution: The coordinate basis vectors are

$$\{\partial_t, \partial_r, \partial_\phi, \partial_z\}.$$

Using the metric,

$$g_{\mu\nu} = \text{diag}(-1, 1, r^2, 1),$$

their squared norms are:

$$\begin{aligned}\|\partial_t\|^2 &= g_{tt} = -1, \\ \|\partial_r\|^2 &= g_{rr} = 1, \\ \|\partial_\phi\|^2 &= g_{\phi\phi} = r^2, \\ \|\partial_z\|^2 &= g_{zz} = 1.\end{aligned}$$

Because $\|\partial_\phi\|^2 \neq 1$, the basis vectors do not all have unit magnitude. Therefore, the coordinate basis is *not orthonormal*, despite the fact that spacetime itself is flat. Flatness refers to curvature, not to the form of the metric components in a chosen coordinate system.

- (b) (4 points) Orthonormal Basis

Define a new set of basis vectors

$$\hat{e}_{\hat{t}} = \partial_t, \quad \hat{e}_{\hat{r}} = \partial_r, \quad \hat{e}_{\hat{\phi}} = \frac{1}{r}\partial_\phi, \quad \hat{e}_{\hat{z}} = \partial_z.$$

1. Show that this basis is orthonormal.
2. Explain the physical meaning of this basis.

Solution: We compute the squared norms:

$$\begin{aligned}\|\hat{e}_t\|^2 &= g_{tt} = -1, \\ \|\hat{e}_r\|^2 &= g_{rr} = 1, \\ \|\hat{e}_\phi\|^2 &= g_{\phi\phi} \left(\frac{1}{r}\right)^2 = r^2 \cdot \frac{1}{r^2} = 1, \\ \|\hat{e}_z\|^2 &= g_{zz} = 1.\end{aligned}$$

All basis vectors have unit magnitude and are mutually orthogonal, so the basis is orthonormal.

Physically, this basis corresponds to the axes defined by an observer's local rulers and clocks. In this basis, the metric takes the Minkowski form $\eta_{\hat{\mu}\hat{\nu}} = \text{diag}(-1, 1, 1, 1)$, so distances and times measured in this frame correspond directly to physical measurements.

(c) (6 points) Vector Components in Different Bases

Consider the vector

$$\vec{V} = V^r \partial_r + V^\phi \partial_\phi$$

with

$$V^r = 0, \quad V^\phi = \omega,$$

where ω is a constant.

1. Compute the magnitude of $\vec{V}$ using the metric.
2. Find the components of $\vec{V}$ in the orthonormal basis.
3. Explain why the orthonormal components correspond to physical measurements.

Solution: The squared magnitude is

$$|\vec{V}|^2 = g_{\phi\phi}(V^\phi)^2 = r^2\omega^2.$$

Thus,

$$|\vec{V}| = r\omega.$$

To find orthonormal components, write

$$\vec{V} = V^{\hat{\phi}} \hat{e}_{\hat{\phi}} = V^{\hat{\phi}} \frac{1}{r} \partial_\phi.$$

Matching coefficients,

$$V^{\hat{\phi}} = r\omega.$$

The orthonormal component $V^{\hat{\phi}}$ corresponds directly to the tangential speed measured by a local observer. Coordinate components such as V^ϕ depend on the coordinate choice and do not directly represent measured quantities.

(d) (2 points) An observer at fixed r and z moves in the ϕ direction with angular velocity ω .

1. Which basis corresponds to what the observer measures?
2. Why can coordinate components be misleading?

Solution: The observer's measurements correspond to the orthonormal basis. This basis is constructed so that each basis vector has unit length and corresponds to physical rulers and clocks.

Coordinate components are misleading because coordinates are labels, not measurements. A coordinate difference $\Delta\phi$ does not correspond to a fixed physical distance unless it is converted using the metric.

2. (10 points) **Euler-Lagrange** Let

$$L = \frac{1}{2} m g_{ij}(x) \dot{x}^i \dot{x}^j. \quad (1)$$

Prove that the Euler-Lagrange equations can be written as

$$\ddot{x}^i + \Gamma_{jk}^i \dot{x}^j \dot{x}^k = 0, \quad (2)$$

where $\Gamma_{jk}^i(x) = \frac{1}{2} g^{i\ell} (\partial_k g_{\ell j} + \partial_j g_{\ell k} - \partial_\ell g_{jk})$.

Solution: The Euler-Lagrange equations state that

$$\frac{\partial L}{\partial x^i} - \frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{x}^i} \right) = 0. \quad (3)$$

For the Lagrangian given in the problem, we find that

$$\frac{\partial L}{\partial x^i} = \frac{1}{2} m \left(\frac{\partial g_{kj}}{\partial x^i} \dot{x}^k \dot{x}^j \right), \quad (4)$$

and

$$\begin{aligned} \frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{x}^i} \right) &= \frac{\partial}{\partial t} [m g_{ij} \dot{x}^j] \\ &= m \left[\frac{\partial g_{ij}}{\partial t} \dot{x}^j + g_{ij} \ddot{x}^j \right] \\ &= m \left[\frac{\partial g_{ij}}{\partial x^k} \dot{x}^k \dot{x}^j + g_{ij} \ddot{x}^j \right]. \end{aligned} \quad (5)$$

Plugging this into the Euler-Lagrange equations, we have

$$\frac{\partial g_{ij}}{\partial x^k} \dot{x}^k \dot{x}^j + g_{ij} \ddot{x}^j + \frac{1}{2} \frac{\partial g_{kj}}{\partial x^i} \dot{x}^k \dot{x}^j = 0 \quad (6)$$

$$= \left(\frac{\partial g_{ij}}{\partial x^k} - \frac{1}{2} \frac{\partial g_{kj}}{\partial x^i} \right) \dot{x}^k \dot{x}^j + g_{ij} \ddot{x}^j . \quad (7)$$

Note that the first term is symmetric with respect to k, j so we can re-write the last line as

$$\frac{1}{2} \left(\frac{\partial g_{ij}}{\partial x^k} + \frac{\partial g_{ik}}{\partial x^j} - \frac{\partial g_{kj}}{\partial x^i} \right) \dot{x}^k \dot{x}^j + g_{ij} \ddot{x}^j = 0 . \quad (8)$$

We can then multiply both sides by the inverse metric $g^{\ell i}$ and note that $g^{\ell i} g_{ij} = \delta_j^\ell$ to get

$$\begin{aligned} & \frac{1}{2} g^{\ell i} \left(\frac{\partial g_{ij}}{\partial x^k} + \frac{\partial g_{ik}}{\partial x^j} - \frac{\partial g_{kj}}{\partial x^i} \right) \dot{x}^k \dot{x}^j + \ddot{x}^\ell \\ &= \frac{1}{2} g^{i\ell} \left(\frac{\partial g_{\ell j}}{\partial x^k} + \frac{\partial g_{\ell k}}{\partial x^j} - \frac{\partial g_{kj}}{\partial x^\ell} \right) \dot{x}^k \dot{x}^j + \ddot{x}^i \\ &= \Gamma_{jk}^i \dot{x}^k \dot{x}^j + \ddot{x}^i \\ &= 0 , \end{aligned} \quad (9)$$

where in the second line we have relabeled $i \leftrightarrow \ell$.

3. **Practice with Christoffel symbols** Consider the geometry of a 2-sphere of radius R . We define the usual coordinate system on the sphere with coordinate θ as the co-latitude (i.e. $\pi/2$ -latitude) and ϕ as the longitude.

(a) (4 points) Show, using a geometric argument, that the line element is

$$ds^2 = R^2 d\theta^2 + R^2 \sin^2 \theta d\phi^2 . \quad (10)$$

Solution: Suppose I have two points A and B infinitesimally far apart on the same longitude. The distance between them is given by $Rd\theta$. Now consider a third point, C , which is on the same latitude as B , at a distance $R \sin \theta d\phi$ from B . By the Pythagorean theorem, the distance between A and C is then $ds^2 = R^2 d\theta^2 + R^2 \sin^2 \theta d\phi^2$.

- (b) (12 points) Find the metric tensor components $g_{\mu\nu}$ and the Christoffel symbols $\Gamma_{\alpha\beta}^\mu$ on the sphere.

Solution: Reading off of the line-element above, we see that the metric tensor is given by

$$g_{\mu\nu} = \begin{pmatrix} R^2 & 0 \\ 0 & R^2 \sin^2 \theta \end{pmatrix} \quad (11)$$

where the indices $\mu, \nu = \theta, \phi$. The only non-zero metric component derivative is $\partial_\theta g_{\phi\phi} = 2R^2 \sin \theta \cos \theta$, so that the only non-zero Christoffel symbols are

$$\Gamma_{\theta\phi\phi} = -R^2 \sin \theta \cos \theta, \Gamma_{\phi\theta\phi} = \Gamma_{\phi\phi\theta} = R^2 \sin \theta \cos \theta, \quad (12)$$

and the other 5 are zero. We can use the inverse metric

$$g^{\mu\nu} = \begin{pmatrix} R^{-2} & 0 \\ 0 & R^{-2} \csc^2 \theta \end{pmatrix} \quad (13)$$

to raise the first indices on the Christoffel symbols to get

$$\Gamma_{\phi\phi}^\theta = -\sin \theta \cos \theta, \Gamma_{\theta\phi}^\phi = \Gamma_{\phi\theta}^\phi = \cot \theta, \quad (14)$$

with the rest zero.

4. (★) Geodesics on the sphere

- (a) (4 points) Write the geodesic equations on the sphere in the form of $\ddot{\theta} = f(\theta, \phi, \dot{\theta}, \dot{\phi})$ and $\ddot{\phi} = g(\theta, \phi, \dot{\theta}, \dot{\phi})$, where the dot indicates a derivative with respect to arc-length.

Solution: From the previous problem, we know that there are only three non-zero Christoffel symbols and that the geodesic equation is $\ddot{x}^\mu = -\Gamma_{\alpha\beta}^\mu \dot{x}^\alpha \dot{x}^\beta$. Writing down the non-zero contributions, we get

$$\ddot{\theta} = \sin \theta \cos \theta \dot{\phi}^2, \ddot{\phi} = -2 \cos \theta \dot{\theta} \dot{\phi}, \quad (15)$$

where $\Gamma_{\phi\phi}^\theta$ contributes to the first equation and $\Gamma_{\theta\phi}^\phi = \Gamma_{\phi\theta}^\phi$ contribute to the second. Note that the factor of 2 results from having both Christoffel symbols.

- (b) (5 points) Show from these equations that the quantities

$$\tilde{L} = R^2 \sin^2 \theta \dot{\phi} \text{ and } \mathbf{u} \cdot \mathbf{u} = R^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) \quad (16)$$

are conserved. What is the interpretation of $\tilde{L}$?

Solution: Conservation implies that $d\tilde{L}/s = d(\mathbf{u} \cdot \mathbf{u})/ds = 0$. We can use the

results of the previous part to see that

$$\begin{aligned}\frac{d\tilde{L}}{ds} &= \frac{d}{ds}(R^2 \sin^2 \theta \dot{\phi}) \\ &= 2R^2 \sin \theta \cos \theta \dot{\theta} \dot{\phi} + R^2 \sin^2 \theta \ddot{\phi} \\ &= 2R^2 \sin \theta \cos \theta \dot{\theta} \dot{\phi} - R^2 \sin^2 \theta (\cos \theta \ddot{\theta} \dot{\phi}) = 0,\end{aligned}\tag{17}$$

and

$$\begin{aligned}\frac{d(\mathbf{u} \cdot \mathbf{u})}{ds} &= \frac{d}{ds}[R^2(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2)] \\ &= R^2[2\dot{\theta}\ddot{\theta} + 2\sin \theta \cos \theta \dot{\theta} \dot{\phi}^2 + 2\sin^2 \theta \dot{\phi} \ddot{\phi}] \\ &= R^2[2\dot{\theta}(\sin \theta \cos \theta \dot{\phi}^2) + 2\sin \theta \cos \theta \dot{\theta} \dot{\phi}^2 - 2\sin^2 \theta \dot{\phi}(2\cot \theta \dot{\theta} \dot{\phi})] \\ &= R^2(2 + 2 - 4)\sin \theta \cos \theta \dot{\theta} \dot{\phi}^2 = 0.\end{aligned}\tag{18}$$

$\tilde{L}$ is the conjugate momentum to ϕ , i.e. angular momentum around the z -axis. This follows from the observation that the metric is independent of ϕ and so by Noether's theorem the conjugate momentum $u_\phi = g_{\mu\phi}u^\mu$ should be conserved. Since $g_{\phi\phi} = R^2 \sin^2 \theta$ and $g_{\phi\theta} = 0$, we see that $u_\phi = R^2 \sin^2 \theta \dot{\phi} = R^2 \sin^2 \theta \dot{\phi}$, which is what we have defined as $\tilde{L}$.

For the rest of the problem, we will take the normalization that $\mathbf{u} \cdot \mathbf{u} = 1$.

- (c) (5 points) Use the result of part (b) to write an equation for $\dot{\theta}$ in terms of θ and $\tilde{L}$. Use this to show that s can be written as

$$s = \pm R \int \frac{d\theta}{\sqrt{1 - \frac{\tilde{L}^2}{R^2 \sin^2 \theta}}}.\tag{19}$$

Solution: Starting with $R^2(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) = 1$, we see that

$$\dot{\theta} = \pm \sqrt{\frac{1}{R^2} - \sin^2 \theta \dot{\phi}^2} = \pm \sqrt{\frac{1}{R^2} - \sin^2 \theta \left(\frac{\tilde{L}}{R^2 \sin^2 \theta} \right)^2} = \pm \frac{1}{R} \sqrt{1 - \frac{\tilde{L}^2}{R^2 \sin^2 \theta}}.\tag{20}$$

Recognizing that $\dot{\theta} = d\theta/ds$, we find that

$$s = \int \frac{d\theta}{\dot{\theta}} = \pm R \int \frac{d\theta}{\sqrt{1 - \frac{\tilde{L}^2}{R^2 \sin^2 \theta}}}.\tag{21}$$

- (d) (6 points) Evaluate this integral and show that

$$\cos \theta = \sin I \cos \frac{s - C}{R},\tag{22}$$

where $\tilde{L} = R \cos I$ and C is some constant. *hint: first do the substitution $z = R \cos \theta$ and then $z = R \sin I \cos \psi$.*

Solution:

$$\begin{aligned}
 s &= \pm R \int \frac{d\theta}{\sqrt{1 - \frac{\tilde{L}^2}{R^2 \sin^2 \theta}}} \\
 &= \mp R \int \frac{dz / \sqrt{R^2 - z^2}}{\sqrt{1 - \frac{\tilde{L}^2}{R^2 - z^2}}} \\
 &= \mp R \int \frac{dz}{\sqrt{R^2 - z^2 - \tilde{L}^2}} \\
 &= \mp R \int \frac{dz}{\sqrt{R^2 \sin^2 I - z^2}} \\
 &= \pm R \int \frac{R \sin I \sin \psi d\psi}{\sqrt{R^2 \sin^2 I - R^2 \sin^2 I \cos \psi}} \\
 &= \pm R \int d\psi = \pm R \psi + C.
 \end{aligned} \tag{23}$$

Therefore, we have $\psi = \pm(s - C)/R$ and

$$\cos \theta = \frac{z}{R} = \sin I \cos \psi = \sin I \cos \frac{s - C}{R}. \tag{24}$$

Note that we can drop the $\pm$ since the cosine function is even.

- (e) (4 points) Explain, based on your 3D Euclidean intuition, why a particle moving at constant speed along a great circle on a sphere should exhibit sinusoidal motion on the z -axis. What are the interpretations of z , I , and ψ in this picture?

Solution: In uniform circular motion in 2D, the coordinates are usually described as $x = R \cos(s/R)$ and $y = R \sin(s/R)$, where s is path length. Here the circle is tilted, but after a linear coordinate transformation the coordinates of a particle should be linear combinations of $\cos(s/R)$ and $\sin(s/R)$. Of course, any such linear combination can be written in terms of $\cos[(s - C)/R]$ for some C using the sum-of-angles formula:

$$\cos \frac{s - C}{R} = \cos \frac{C}{R} \cos \frac{s}{R} + \sin \frac{C}{R} \sin \frac{s}{R}, \tag{25}$$

meaning we can change the phase so as to only have a cosine term. The 3D physical interpretation of $z = R \cos \theta$ is of course that it is the z -coordinate of the particle. Here I is the inclination of the great circle (the latitude $\pi/2 - \theta$ varies between $\pm I$). If we describe in 3D $\mathcal{O}$ as the center of the sphere, $\mathcal{A}$ as

the point of most northerly latitude, and $\mathcal{P}$ as the point on the particle's path, then ψ can be interpreted as the 3D angle $\angle \mathcal{AOP}$, because the particle is at $\mathcal{A}$ at $s = C$, and then at general s the arclength from $\mathcal{A}$ to $\mathcal{P}$ is $s - C$. It follows that the angle is $(s - C)/R$.